

Reducing algebraic fractions to a common denominator

Factorise every denominator, take each factor to its highest power, and multiply each fraction up.

LESSON 114–115

2 × 40 MIN

MATEMATIKA 7, §51

S8 7

LESSON CLOCK

10'

25'

25'

20'

5'

The rule that makes it legal	10 min
Building the denominator	25 min
The extra factors	25 min
Denominators that differ in sign	20 min
Homework	5 min

LEARNING OBJECTIVES

- Find the lowest common denominator of two algebraic fractions.
- Determine the extra factor each fraction must be multiplied by.
- Multiply numerator and denominator by the same expression without changing the value.
- Handle denominators that differ only in sign.

TEXTBOOK

National	pp. 335–341
Cambridge	Stage 8, pp. 66–78
Workbook	Exercise 7.2

01

Explanation

The rule that makes it legal

Multiplying the numerator and the denominator by the same non-zero expression does not change the value of a fraction — it only changes how it is written.

$$\frac{P}{Q} = \frac{P \cdot R}{Q \cdot R} \text{ for } R \neq 0$$

FRACTION	MULTIPLIED BY	NEW FORM
$\frac{1}{x}$	$x + 1$	$\frac{x + 1}{x(x + 1)}$
$\frac{2}{x - 3}$	$x + 3$	$\frac{2(x + 3)}{x^2 - 9}$
$\frac{3}{a}$	2	$\frac{6}{2a}$

CANCELLING AND BUILDING ARE THE SAME RULE

Last lesson divided both parts by a common factor; this lesson multiplies both parts by an extra one. It is one rule read in two directions.

Building the denominator

Factorise every denominator, then take each distinct factor to the highest power in which it appears anywhere.

DENOMINATORS	FACTORISED	LOWEST COMMON DENOMINATOR
$6x, 8x^2$	$2 \cdot 3 \cdot x, 2^3 \cdot x^2$	$24x^2$
$x - 3, x + 3$	already factorised	$(x - 3)(x + 3)$
$x^2 - 9, x + 3$	$(x - 3)(x + 3), x + 3$	$(x - 3)(x + 3)$
$x^2 + 2x + 1, x^2 - 1$	$(x + 1)^2, (x - 1)(x + 1)$	$(x + 1)^2(x - 1)$

MULTIPLYING THE DENOMINATORS TOGETHER IS NOT WRONG, ONLY WASTEFUL

For $x^2 - 9$ and $x + 3$ the product is $(x - 3)(x + 3)^2$, which works but leaves a factor to cancel at the end. Factorising first gives the lowest one and half the arithmetic.

The extra factors

Each fraction is multiplied by whatever its own denominator is missing.

FRACTION	DENOMINATOR	EXTRA FACTOR	NEW NUMERATOR
$\frac{1}{x-3}$	$x-3$	$x+3$	$x+3$
$\frac{2}{x+3}$	$x+3$	$x-3$	$2x-6$
$\frac{5}{6x}$	$6x$	$4x$	$20x$
$\frac{3}{8x^2}$	$8x^2$	3	9

MULTIPLY THE NUMERATOR OUT

$\frac{2}{x+3}$ becomes $\frac{2x-6}{(x-3)(x+3)}$, not $\frac{2(x-3)}{\dots}$ left unexpanded — the next lesson has to add these numerators, and an unexpanded bracket with a minus in front is where the sign errors start.

Denominators that differ in sign

$3-x$ and $x-3$ are opposites, so one sign change makes them equal.

$\frac{1}{3-x} = -\frac{1}{x-3}$

PAIR	REWRITTEN	COMMON DENOMINATOR
$x-3, 3-x$	$3-x = -(x-3)$	$x-3$
$a-b, b-a$	$b-a = -(a-b)$	$a-b$
$2x-1, 1-2x$	$1-2x = -(2x-1)$	$2x-1$

TREATING THEM AS DIFFERENT FACTORS DOUBLES THE WORK

Building $(x-3)(3-x)$ as a common denominator is legal but pointless — and it hides the fact that the answer will simplify. Take the minus out in front instead.

02 Worked examples

EXAMPLE 1 Reduce $\frac{1}{x-3}$ and $\frac{2}{x+3}$ to a common denominator.

① Neither denominator factorises further.

② The LCD is $(x-3)(x+3) = x^2 - 9$.

③ $\frac{1}{x-3} = \frac{x+3}{x^2-9}$
Extra factor $x+3$.

④ $\frac{2}{x+3} = \frac{2x-6}{x^2-9}$
Extra factor $x-3$.

ANSWER

$$\frac{x+3}{x^2-9} \text{ and } \frac{2x-6}{x^2-9}$$

EXAMPLE 2 Find the lowest common denominator of $\frac{1}{x^2+2x+1}$ and $\frac{1}{x^2-1}$.

① $x^2 + 2x + 1 = (x+1)^2$

② $x^2 - 1 = (x-1)(x+1)$

③ Take $(x+1)$ to its highest power and $(x-1)$ once.

④ LCD = $(x+1)^2(x-1)$.

ANSWER

$$(x+1)^2(x-1)$$

EXAMPLE 3 Reduce $\frac{1}{x-3}$ and $\frac{2}{3-x}$ to a common denominator.

① $3 - x = -(x - 3)$

② $\frac{2}{3-x} = -\frac{2}{x-3}$

③ The common denominator is just $x - 3$.

④ The fractions are $\frac{1}{x-3}$ and $\frac{-2}{x-3}$.

ANSWER

Common denominator $x - 3$

03 Interactive model

Set the pair $1/(x - 3)$ and $2/(3 - x)$ before teaching the sign rule; the class builds a monstrous denominator, and the one-line alternative then sells itself.

Assembling the lowest common denominator

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
$6x$	$2 \cdot 3 \cdot x$	$4x$
$8x^2$	$2^3 \cdot x^2$	3

LOWEST COMMON DENOMINATOR

$$24x^2$$

The numerators become $20x$ and 9 .

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Common denominator	Umumiy maxraj	Общий знаменатель
Lowest common denominator	Eng kichik umumiy maxraj	Наименьший общий знаменатель
Extra factor	Qo'shimcha ko'paytuvchi	Дополнительный множитель
To factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Highest power	Eng yuqori daraja	Наивысшая степень
Equivalent fraction	Teng kuchli kasr	Равная дробь
Opposite expression	Qarama-qarshi ifoda	Противоположное выражение
Sign change	Ishorani o'zgartirish	Смена знака

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Multiplying both parts of a fraction by the same expression:

changes its value

keeps its value

makes it zero

is not allowed

The LCD of $6x$ and $8x^2$ is:

$48x^3$

$24x^2$

$14x^2$

$24x$

The LCD of $x - 3$ and $x + 3$ is:

$x^2 - 9$

$x^2 + 9$

$x - 3$

$(x - 3)^2$

The LCD of $x^2 - 9$ and $x + 3$ is:

$3 - x$ equals:

The extra factor for 2 $x + 3$ towards $x^2 - 9$ is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. LCD of $6x$ and $8x^2$

ANSWER $24x^2$

2. LCD of $x - 3$ and $x + 3$

ANSWER $x^2 - 9$

3. LCD of $2a$ and $3a$

ANSWER $6a$

4. LCD of x and $x + 1$

ANSWER $x(x + 1)$

5. $\frac{1}{x}$ to denominator $x(x + 1)$

ANSWER $\frac{x + 1}{x(x + 1)}$

6. $\frac{3}{a}$ to denominator $2a$

ANSWER $\frac{6}{2a}$

7. $3 - x$ written with $x - 3$

ANSWER $-(x - 3)$

Medium · the standard question

7 PROBLEMS

1. LCD of $x^2 - 9$ and $x + 3$

ANSWER $x^2 - 9$

2. LCD of $x^2 + 2x + 1$ and $x^2 - 1$

ANSWER $(x + 1)^2(x - 1)$

3. $\frac{2}{x + 3}$ to denominator $x^2 - 9$

ANSWER $\frac{2x - 6}{x^2 - 9}$

4. $\frac{1}{x - 3}$ to denominator $x^2 - 9$

ANSWER $\frac{x + 3}{x^2 - 9}$

5. $\frac{5}{6x}$ to denominator $24x^2$

ANSWER $\frac{20x}{24x^2}$

6. LCD of $x - 3$ and $3 - x$

ANSWER $x - 3$

7. LCD of $a - b$ and $b - a$

ANSWER $a - b$

Hard · stretch and reason

7 PROBLEMS

1. LCD of $x^2 - 4$, $x^2 + 4x + 4$ and $x + 2$

ANSWER $(x - 2)(x + 2)^2$

2. LCD of $x^2 - x$ and $x^2 - 1$

ANSWER $x(x - 1)(x + 1)$

3. $\frac{1}{x^2 - 4}$ to denominator $(x - 2)(x + 2)^2$

ANSWER $\frac{x + 2}{(x - 2)(x + 2)^2}$

4. LCD of $4x^2y$ and $6xy^3$

ANSWER $12x^2y^3$

5. Why is the lowest common denominator preferred?

ANSWER The numbers stay small and nothing has to be cancelled at the end

6. $\frac{2}{3 - x}$ with denominator $x - 3$

ANSWER $\frac{-2}{x - 3}$

7. LCD of $x^2 - 9$, $x^2 - 6x + 9$

ANSWER $(x + 3)(x - 3)^2$

07 Homework

Homework — 5 tasks

Factorise every denominator before you write a single common denominator.

- Find the LCD of $10x$ and $15x^2$.
- Reduce $\frac{1}{x - 5}$ and $\frac{3}{x + 5}$ to a common denominator.
- Find the LCD of $x^2 - 16$ and $x + 4$, and rewrite both fractions.

4. Find the LCD of $x^2 - 25$ and $x^2 + 10x + 25$.

5. Rewrite $\frac{4}{2-x}$ with denominator $x - 2$.